

Project Initialization and Planning Phase

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **Credit Card Approval Prediction using Machine Learning** project is designed to improve the efficiency and accuracy of the credit card approval process by utilizing data-driven decision-making. Traditional approval methods often involve manual verification of applicant information, making the process time-consuming and prone to inconsistencies. The proposed solution employs a trained machine learning model to analyze applicant details and predict whether a credit card application is likely to be approved or rejected.

The system integrates a user-friendly web application with a machine learning model, allowing users to enter their information and receive instant prediction results. This approach assists financial institutions in reducing processing time, improving consistency, and supporting better credit risk assessment.

Project Overview	
Objective	The primary objective of this project is to develop an intelligent Credit Card Approval Prediction System using Machine Learning that can accurately predict the approval status of a credit card application based on the applicant's demographic, financial, and employment information. The system aims to support financial institutions by providing quick, reliable, and data-driven predictions while reducing manual effort in the approval process.
Scope	The scope of the project includes collecting and preprocessing the credit card application dataset, training and evaluating multiple machine learning models, selecting the best-performing model, and integrating it into a Flask-based web application. The application enables users to enter applicant information through a web interface and instantly receive an approval prediction. The project focuses on improving prediction efficiency and providing a user-friendly interface while allowing future enhancements such as cloud deployment, user authentication, and explainable AI techniques.
Problem Statement	
Description	Traditional credit card approval systems rely heavily on manual evaluation of applicant information and predefined rules. This process can be time-consuming, inconsistent, and inefficient when handling a large number of applications. Evaluating multiple financial and demographic factors manually increases the possibility of delays and human errors, making it difficult for financial institutions to provide quick and accurate decisions.
Impact	Implementing an automated machine learning-based credit card approval prediction system significantly improves the efficiency of the approval process. The proposed solution reduces processing time, minimizes manual intervention, improves prediction consistency, supports better credit risk assessment, and enhances the overall customer experience by providing faster approval decisions.
Proposed Solution	
Approach	The proposed solution utilizes a Random Forest Machine Learning model trained on historical credit card application data to predict whether a customer's application is likely to be approved or rejected. The system preprocesses applicant information, converts categorical data into numerical values, and passes the processed

	data to the trained model for prediction. A Flask web application serves as the interface between the user and the machine learning model, enabling real-time prediction through a simple and interactive web interface.
Key Features	Machine Learning-based credit card approval prediction. • User-friendly web application developed using Flask. • Real-time prediction of approval or rejection. • Data preprocessing and feature encoding for accurate predictions. • Responsive interface developed using HTML, CSS, and Bootstrap. • High prediction accuracy using the Random Forest Classifier. • Easy integration of the trained model using Joblib. • Modular architecture that supports future enhancements and deployment.

Resource Requirements :

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i5 / AMD Ryzen 5 or higher
Memory	Memory (RAM)	Minimum 8 GB
Storage	Storage	256 GB SSD or higher
Software		
Frameworks	Web Framework	Flask
Libraries	Machine Learning Libraries	Scikit-learn, Pandas, NumPy
	Visualization Libraries	Matplotlib, Seaborn
Development Environment	IDE	Jupyter Notebook, Visual Studio Code
Data		
Data	Dataset	Credit Card Approval Dataset
	Dataset Format	CSV